

Exercise to laboratory 3B: Topography and β -plan

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1 Theory (12P)

- i. State the necessary assumption that are needed to study Rossby waves. (2P)
- ii. Write down the dispersion relation of Rossby waves, the equation for the group velocity and the phase speed. Discuss the dispersion of Rossby waves. (2P)
- iii. Discuss in your own words the term dispersion. (1P)
- iv. From the equations of questions ii., discuss the role of the wave number k (in relation to the Rossby radius R). Which cases can be distinguished? (3P)
- v. State the equation of $\partial h / \partial t$ with variable depth $D = D_0 + \alpha y$ and state/mark the two additional terms that needs to be implemented. (2P)
- vi. State the relation where α cancels the β -effect. (1P)
- vii. Why do we have to assume reduced gravity in this laboratory? (1P)

2 Experiments

Set up the model as described in the laboratory description for laboratory 3B and follow the instruction for the experiments.

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2.1 The role of wave number (10P)

- i. List 6 values of kR and the corresponding Lw that sample the characteristic shape of the zonal Rossby wave phase c_x and group c_{gx} speed relations (if $Lw = 1.3 \frac{\pi}{k \cdot R} \frac{1}{\sqrt{5}} \frac{\sqrt{g'H}}{f_0}$.) and explain your choice. (1.5P)
- ii. Plot the theoretical zonal phase c_x and group c_{gx} speeds as a function of kR and highlight the values that correspond to your choice of i. (1.5P)
- iii. Choose three kR values that represent the characteristic regimes of the Rossby wave and provide Hovmoeller diagrams at the center of your y-axis. Estimate the numerical zonal phase and group velocity and plot them as lines into your plots. Add the theoretical zonal phase and group speed as lines into the Hovmoeller diagram. (6P)
- iv. Add the numerical estimates of your three characteristic waves into the plot from ii. (1P) (It is fine if you include one final plot for question ii. and iv. together.)

2.2 The role of topography (8P)

- i. Now, fix β and choose a value for kR for your next experiments of a variation in α . Plot Hovmoeller diagrams of at least three representative waves on the α -plane. Add the theoretical zonal phase and group speed to the plot and explain your parameter choice. (4P)
- ii. Plot your waves shown in i. in xy -cross sections at an appropriate time. (1.5P)
- iii. For the settings of two of your α -value experiments (not the cancelling one!), plot the theoretical c_g and c_x as a function of kR and include the numerical estimates of this experiment. Compare to the plot from 2.1.iv. (2.5P)

Good luck!